A120598 concise calculus certificate

VERIFIED

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Definition and first terms

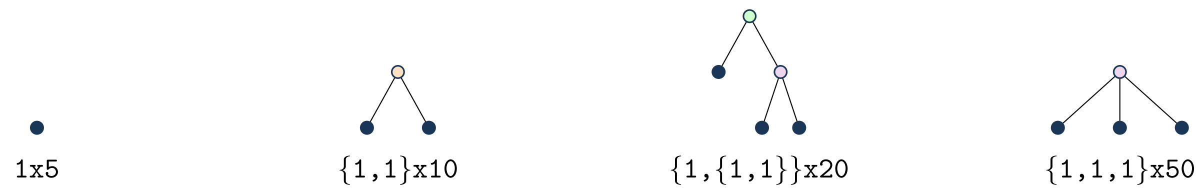
$30\,A(x) = 29 + 125\,x + A(x)^5$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(5)*T(x); T=x+2*T^2+10*T^3+25*T^4+25*T^5`.

$a(0), \ldots, a(7) = 1, 5, 10, 90, 825, 8445, 92820, 1066740.$

Kernel used throughout: $\rho(u) = u\,(1 - 2\,u^1 - 10\,u^2 - 25\,u^3 - 25\,u^4)$, so $x = \rho(T(x))$.

Typogeometry



colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2: 2, Δ_3: 10, Δ_4: 25, Δ_5: 25.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_4) \in \mathbb{Z}^4_{\geq 0} : \sum_{j=1}^{4} j k_j = n - 1\}, \quad K = \sum_{j=1}^{4} k_j,$$

$$a(n) = \frac{5}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n+K-1}{K} \binom{K}{k_1, \ldots, k_4} 2^{k_1} 10^{k_2} 25^{k_3} 25^{k_4}.$$

$$a(n) = \frac{5}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{4} P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-4 coefficient polynomials are embedded in `certificate_payload.json` (source: `release/certificate_payload.json.gz#/objects/p_recurrence`).

Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/certificate_payload.json.gz#/objects/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u) = N(n, u)/\rho(u)^3$, $\deg_u N = 16$.

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 10×10; remainder 4×5, rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.